

PROMETHEAN — Final Submission Packet

Project Information

- **Project title:** Prometheus Studio
- **Team members:** Noah Hicks, Rushabh Kankariya, Vishnu Bala, Yiyang Lu, Mel Wong
- **Selected track:** Track A: Technical Build
- **Repository link:** <https://github.com/RedLynx101/Promethean>
- **5-minute demo video:** <https://drive.google.com/drive/folders/19-IFyg5feKVrce7TfKstvEAJC69OSSJh>

One-paragraph project summary

Promethean Studio is an AI workflow orchestration system that transforms natural-language workflow descriptions into deployable, governed workflow graphs through a four-agent pipeline: Decomposition, System Selection, Orchestration, and Governance. The system is designed to reduce both over-automation and under-automation by assigning each node the right autonomy level on an L0-L5 spectrum, while enforcing human-in-the-loop approval gates between every phase transition. The deliverable includes a visual graph editor for review and correction, full trace capture for reproducibility, scenario-based evaluation with baseline comparisons (Always-L0/L4/L5), and failure analysis with root cause documentation, verified fixes, and clearly named residual limitations.

Required Submission Materials

- **Final report:** file included in packet (repo source: `phase-3/final_report.md`)
 - **Architecture diagram:** file included in packet (repo source: `phase-2/01-architecture-diagram.md`)
 - **Screenshot index:** file included in packet (repo source: `phase-3/screenshots/screenshot_index.md`)
 - **Evaluation summary:** included below (repo sources: `phase-3/final_report.md`, `phase-3/evidence/`)
 - **List of submitted files and folders:** included below
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Evaluation Summary

Scope and method

Evaluation was run on eight total executions spanning five scenarios: TC-01 (CRM happy path), TC-02 (cybersecurity incident response), TC-03 (deterministic CSV validation), TC-04 (ambiguous input), and TC-05a/05b/05c/05d (boundary and adversarial cases). Runs were executed against the live pipeline with full JSON traces and post-run metric extraction, not synthetic mocks.

Quantitative outcomes

- **Pipeline completion:** 8/8 runs completed successfully to terminal outcome (including expected 400 boundary rejection for TC-05a post-fix).
- **Schema validation:** 100% pass rate across all four agents and all completed scenarios.
- **Latency profile:** approximately 96s median total agent latency, with TC-05c worst-case at approximately 161s after long-input fix.
- **Cost profile:** Promethean consistently outperformed the Always-L4 and Always-L5 baselines; reported savings vs Always-L4 ranged from roughly 4.9x to 225x across scenarios.
- **Failure handling maturity:** four concrete failure cases were documented; three were fixed and re-verified with before/after traces, one remains open as an explicit limitation.

Nuanced interpretation

- The strongest evidence is not just aggregate pass rates, but pre-fix vs post-fix trace pairs for FC-01 and FC-02 that show concrete behavioral correction under the same workflow families.

- TC-04 and TC-05d demonstrate that the team did not collapse all uncertainty into “pass” claims; overconfidence on vague input and non-workflow prompt handling are explicitly surfaced as residual risks.
- Baseline comparisons are methodologically useful for cost and calibration framing, but should be interpreted as design-time comparisons, not production execution benchmarks.

Primary evidence references

- **Repo source paths:**
 - phase-3/evidence/evaluation_results.csv
 - phase-3/evidence/failure_log.md
 - phase-3/evidence/version_notes.md
 - phase-3/evidence/baseline_comparison_crm.csv
 - phase-3/evidence/baseline_comparison_crm.md
 - phase-3/traces/
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List of Submitted Files and Folders

Phase 3 deliverable package

- **Repo source paths:**
- phase-3/final_report.md (file included in packet)
- phase-2/01-architecture-diagram.md (file included in packet)
- phase-3/screenshots/screenshot_index.md (file included in packet)
- phase-3/evidence/
 - test_cases.csv
 - test_cases.md
 - evaluation_results.csv
 - failure_log.md
 - version_notes.md
 - baseline_comparison_crm.csv
 - baseline_comparison_crm.md
- phase-3/screenshots/
 - 04_intake_review_gate.png
 - 05_agent_output_detail.png
 - 06_approval_gate_inter_agent.png
 - 07_govern_gate.png
 - 08_deployed_gate.png
- phase-3/traces/
- phase-3/reflections/
 - melissa_wong.md
 - noah_hicks.md
 - rushabh_kankariya.md
 - vishnu_bala.md
 - yiying_lu.md